

# **ENGEN370 Environmental Impact Assessment (Five)**

# Guide to Preparing an EIA

DOC has a simple step by step guide which can provide a basis or starting point for preparing your EIA for your project. See the reading list.

Step 1 Describe the environment

Step 2 Scope out the potential effects of your project

Step 3 Identify effects of your project and activities

Step 4 Describe any measures you can take to eliminate, avoid, remedy or mitigate the identified impacts

Step 5 Identify and evaluate alternatives

Step 6 Outline any monitoring programme you have developed

Plus consider risk and resilience

# Guide to Preparing an EIA

## Step 1 Describe the environment

- Describe the existing natural environment
  - Air
  - Water
  - Soil
  - Ecosystems
- Identify significant natural features of the landscape.
  - Rivers, lakes, springs, falls
  - Woodland, forest, alpine, islands, beaches
  - Mountains, fjords, glaciers

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Step 2 Scope out the potential effects of your project.

For example for a construction type project:

- Transport and Access
- Lead to any erosion or slippage?
- Construction
- Long term adverse effects such as those resulting from the removal of rock, soil, or vegetation
- Site Restoration
- Water/Soil
- Effluent/Waste Disposal - especially sewerage.
- Hazardous Substances/Contaminants
- Ongoing operation and monitoring of facility

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## Step 3 Identify effects of your project and activities

- Identify any and all direct, indirect and cumulative effects where possible
- any positive or beneficial as well as potential adverse effects on the environment

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Step 4 Describe any measures you can take to:

- Eliminate
- Avoid
- Remedy
- Mitigate

the identified impacts

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Step 5 Identify and evaluate alternatives such as:

- Alternative designs
- Alternative methods (e.g. construction)
- Alternative locations
- Alternative technologies
- Alternative timescales

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Step 6 Outline any

- Monitoring
- rehabilitation programme you may have identified

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The Guide includes an effect identification and mitigation checklist

The checklist comprises a table summarising :

- Values
- Potential adverse effects on those values
- Mitigation measures that can be taken

A useful tool to help the process along and get you thinking

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Values include:

- Land
- Water
- Historic
- Cultural
- Landscape
- Recreation
- Infrastructure
- Others (Air? Climate?)

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Potential adverse effects include:

- Damage to biodiversity
- Destruction of habitat
- Discharge of pollutants
- Noise
- Employment/income
- Restricting access
- Erosion, scour, deposition
- Impact on water (level, pathway, availability)
- Impact on historic/culturally important sites
- Safety
- Economic impact
- Communities
- Cumulative effects over time

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Consider mitigation measures you can undertake to :

- Reduce the impact
- Avoid the impact
- Eliminate the impact
- Mitigate the impact

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Mitigation measures include for example:

- Alternative designs
- Alternative technologies
- Alternative locations
- Risk assessment and risk management
- Monitoring programmes
- Post project rehabilitation
- Consultation with stakeholders e.g. Iwi

# Short Exercise

- The Waikato Expressway upgrade is a major civil engineering project
- 2009 to 2021 \$100+ million
- Comprises some 7 sections (4 currently complete)
- Stretching some 116km from south of Cambridge to the Auckland Southern Motorway
- Passes through/around Hamilton



Source: <https://www.nzta.govt.nz/assets/projects/hamilton/docs/po-ster-delivering-the-waikato-expressway.pdf>

## Short Exercise

The table provided (on moodle) lists the following values for consideration:

- Land
- Air
- Water
- Historic
- Social
- Cultural
- Economic
- Landscape
- Recreation/access
- Infrastructure

## Short Exercise

Using columns 2 and 3 of the table provided think about and write down some brief points on each of the following :

- Potential adverse effects on the Values
- Potential mitigation measures that could be taken